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Chapter - 3
Distribution and Habitat Preferences of the
***Mauremys caspica* Gmelin, 1774 in Turkey**

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Chapter - 3

Distribution and Habitat Preferences of the *Mauremys caspica* Gmelin, 1774 in Turkey

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Abstract

Geographical distribution has an importance to contribute the conservation, ecology and evolution of the species. Many species have different evolutionary processes in their distribution area, and these processes were affected by many biotic and abiotic factors. In the present study, we contributed to knowledge of distribution of the Caspian stripe-necked terrapins, *M. caspica* in Turkey using geographical and bioclimatic data. The distribution of species was generally observed to be between the northwestern Black sea region and the northern of the central Anatolia region, between eastern Anatolia and southern Anatolia regions. In addition, our model shows that the species may be distributed in low level in the other parts of Turkey. The environmental variables contributed to highest gain for distribution of *M. caspica* were elevation (21.4%), Bio 12 (Annual Precipitation, 20.5%), Bio 10 (Mean Temperature of Warmest Quarter, 14.8%) and Bio 17 (Precipitation of Driest Quarter, 11.6%). Other variables showed a lower contribution than percentage 35. The final constructed models resulted in good and high AUC with a low standard deviation (0.912 ± 0.082) for *M. caspica*. The high AUC result and lower standard deviation exhibit that potential distribution area of the species was congruent with its habitat preferences inhabited localities.

Keywords: Geography, ecology, caspian stripe-necked terrapins, habitat, potential distribution

Introduction

Ecological niche modeling is very useful technique to learn distribution of the animal species, and today it plays an important role in many areas of conservation, ecology distribution, geography and evolutionary biology, both because they can fill gaps in knowledge and allow a better estimate of multiple components of species diversity (Guisan and Zimmermann, 2000; Araújo and Guisan, 2006; Phillips *et al.*, 2006; Ashraf *et al.*, 2017). This approach

analyzes the distribution model integrating the distribution data (longitude and altitude) of a species with the bioclimatic variables (abiotic variables such as rainfall and temperature). As a result of the analysis, the model creates a propagation map by revealing the habitat preferences of the species (Guisan and Thuiller, 2005; Araújo and New, 2007).

The genus *Mauremys* (Gray, 1869) have nine recognized species (*Mauremys annamensis* (Siebenrock, 1903), *Mauremys caspica* (Gmelin, 1774), *Mauremys japonica* (Temminck & Schlegel, 1835), *Mauremys leprosa* (Schweigger, 1812), *Mauremys mutica* (Cantor, 1842), *Mauremys nigricans* (Gray, 1834), *Mauremys reevesii* (Gray, 1831), *Mauremys rivulata* (Valenciennes, 1833) and *Mauremys sinensis* (Gray, 1834)) distributed in the Palearctic and oriental zone of the world (<http://reptile-database>). The genus as known stripe-necked terrapins are represented by two species (*M. caspica* and *M. rivulata*) in Turkey (Baran and Atatür, 1998; Fritz *et al.*, 2008; Baran *et al.*, 2012). *M. caspica* is distributed between eastern and southeastern Anatolia and from the middle Anatolian Plateau, while the range of *M. rivulata* distributes from southeastern Europe to the Middle East, including western Anatolia and the coastal strip of Syria into Lebanon, Israel, and Jordan (Haas 1952; Mertens and Wermuth 1960; Wermuth and Mertens 1961; Başoğlu and Baran 1977; Iverson 1986; Ernst and Barbour 1989; Fritz and Freytag 1993; Baran and Atatür 1998; Taşkavak *et al.* 1998; Wischuf and Fritz 2001; Wischuf and Busack 2001).

M. caspica was first described by Gmelin (1774) from Pyrsagat in Transcaucasia. The first report was given by Boulenger from Ankara for the species in Turkey (Ayaz *et al.*, 2006). After the first record of the species from Turkey, the study performed on the species was on the distribution, ecology and morphology (Bird, 1936; Bodenheimer, 1944; Mertens 1946; Loveridge and Williams 1957; Wermuth and Mertens 1961; Eiselt and Spitzenberger, 1967; Wermuth and Mertens, 1977; Başoğlu and Baran, 1977; Busack and Ernst, 1980; Çevik, 1982; Fritz and Freytag, 1993; Tok, 1993; Fritz and Wischuf, 1995; Fritz and Wischuf, 1997; Baran and Atatür, 1998).

M. caspica is one of the medium-sized aquatic species, occurring in a doubly disjunct range in Anatolia. The carapace of the species is about 25 cm and it fused with the plastron. The lake, river, ditch and fresh water-like places e upper part of the back shell is gray, with dark or black spots at the bottom. Yellow lines are available on the black, gray-colored, head, neck, feet and tail. The habitat of species comprises lake, river, ditch and fresh water-like places (Baran and Atatür, 1998; Baran *et al.*, 2012).

Many studies on the potential distribution of amphibians and reptiles have been performed by many researchers in the recent years (Fattahi *et al.*, 2014; Gül, 2015; Gül, *et al.*, 2015; Yousefkani *et al.*, 2016) since their distribution in the areas and habitat preferences of these animals are important for their conservation and ecology (Kumar and Stohlgren, 2009; Gül, 2013). According to literature, the species distributed in central, east and southeast Anatolia (Baran *et al.*, 2012). The present study was to give first distributional data from the eastern black sea region. Also, this is the first study presenting habitat and climatic preference using bioclimatic variables as well as distribution.

Materials and Methods

A total of 42 locality records were collected from literature and field study for *M. caspica* (Clark and Clark 1973; Teynie, 1991; Fritz and Freytag, 1993; Gramentz, 1993; Schweiger 1994; Taşkavak *et al.* 1998; Ayaz *et al.* 2008; Fritz *et al.* 2008) (Table 1).

Table 1: The locality data of 6 lineages of *M. caspica* in Turkey

Latitude, Longitude	References	Locality
40.183150, 40.305717	This study	Gez village, Bayburt
38.699008, 41.706861	Schweiger 1994	Hasköy, Muş
37.647991, 38.942630	Schweiger 1994	Hilvan, Şanlıurfa
37.596783, 38.856735	Schweiger 1994	Hilvan, Şanlıurfa
37.772354, 38.981883	Schweiger 1994	Çaylarbaşı-Hilvan, Şanlıurfa
39.975190, 41.670002	Schweiger 1994	Pasinler, Erzurum
40.107749, 42.426797	Schweiger 1994	Horasan, Erzurum
40.151069, 42.585503	Schweiger 1994	Karakurt, Kars
40.151651, 43.325081	Schweiger 1994	Aydıncavak, Kağızman, Kars
40.116359, 43.121844	Schweiger 1994	Kağızman, Kars
40.658293, 35.836176	Schweiger 1994	Amasya
40.387079, 36.089725	Schweiger 1994	Turhal, Tokat
41.051231, 35.599111	Schweiger 1994	Havza, Samsun
40.790147, 35.557220	Schweiger 1994	Merzifon, Amasya
41.111999, 34.622249	Schweiger 1994	Tosya, Kastamonu
38.882813, 35.753789	Schweiger 1994	Bünyan, Kayseri
40.985365, 34.099142	Schmidtler, 1986	Tosya, Kastamonu
38.402565, 34.137147	Fritz and Freytag, 1993	Aksaray
39.729322, 33.484152	Teynie, 1991	Elmadag-Kırıkkale arası
39.844586, 37.096129	Fritz and Freytag, 1993	Sivas
40.050050, 37.333929	Teynie, 1991	Hafik, Sivas

38.046412, 39.284571	Teynie, 1991	Çermik-Siverek arası
37.962877, 42.097853	Teynie, 1991	Baykan, Siirt
38.162183, 41.203344	Clark and Clark 1973	Silvan, Diyarbakır
37.143802, 39.878337	Teynie, 1991	Viranşehir, Şanlıurfa
37.246827, 37.865785	Gramentz, 1993	Halfeti, Şanlıurfa
36.086501, 36.324232	Teynie, 1991	Altınözü, Hatay
37.825820, 38.646793	Ayaz <i>et al.</i> 2008; Fritz <i>et al.</i> 2008	Adıyaman
38.235584, 33.560996	Ayaz <i>et al.</i> 2008; Fritz <i>et al.</i> 2008	Sultanhanı, Aksaray
37.911571, 40.250715	Ayaz <i>et al.</i> 2008; Fritz <i>et al.</i> 2008	Diyarbakır
39.533615, 31.749925	Ayaz <i>et al.</i> 2008; Fritz <i>et al.</i> 2008	Oğlakçı, Sivrihisar, Eskişehir
38.887063, 35.744251	Ayaz <i>et al.</i> 2008; Fritz <i>et al.</i> 2008	Sarımsak Barajı, Kayseri
37.070424, 37.872430	Ayaz <i>et al.</i> 2008; Fritz <i>et al.</i> 2008	Birecik, Şanlıurfa
37.024870, 37.974139	Ayaz <i>et al.</i> 2008; Fritz <i>et al.</i> 2008	Birecik, Şanlıurfa
37.657712, 39.080402	Ayaz <i>et al.</i> 2008; Fritz <i>et al.</i> 2008	Çaylarbaşı-Hilvan, Şanlıurfa
41.315907, 34.924872	Ayaz <i>et al.</i> 2008; Fritz <i>et al.</i> 2008	Boyabat, Sinop
40.568173, 36.923864	Ayaz <i>et al.</i> 2008; Fritz <i>et al.</i> 2008	Sarıyazı, Niksar, Tokat
38.536481, 43.416644	Ayaz <i>et al.</i> 2008; Fritz <i>et al.</i> 2008	Van
39.382171, 33.429870	Taşkavak <i>et al.</i> 1998	Kesikköprü, Ankara
38.887947, 35.743930	Taşkavak <i>et al.</i> 1998	Sarımsaklı Göleti, Kayseri
39.824605, 32.827256	Taşkavak <i>et al.</i> 1998	Eymir Gölü, Ankara
38.756526, 35.420298	Taşkavak <i>et al.</i> 1998	Şeker Gölü, Kayseri

For species distribution analysis, firstly 19 bioclimatic variables were downloaded from Global Climate Data (Hijmans *et al.*, 2005, available at www.worldclim.org) (Table 2). These data were generated from global esri grids in the highest resolution 30 arc-seconds (approximately 1 km) for current conditions (~1950-2000). Each bioclimatic variable was limited the land border of Turkey using by Arc Toolbox (extract by mask) in ArcGIS 10.3 software version. For the elevation, SRTM30 dataset from CGIAR SRTM aggregated to 30 seconds were used (available at <http://srtm.csi.cgiar.org/>). To eliminate the negative effect may be resulted from other environmental

variables, Pearson correlation coefficient ($0.75 < r < -0.75$) was performed using ENM Tools 1.4 (Warren *et al.*, 2010) for all variables, and highly correlated variables were left out of analyze for distribution modeling of *M. caspica*. After the correlation analyze, we chose nine bioclimatic variables (Bio 12, Bio 10, Bio 17, Bio 19, Bio 7 Bio 13, Bio 4, Bio 2 and Bio 14) and elevation.

Table 2: Summary of environmental variables from the Worldclim data set and contributions of the environmental variables

Bioclimatic Variable	Description
Bio 1	Annual Mean Temperature
Bio 2	Mean Diurnal Range (Mean of monthly (max temp-min temp))
Bio 3	Isothermality (BIO2/BIO7) (*100)
Bio 4	Temperature Seasonality (standard deviation *100)
Bio 5	Max Temperature of Warmest Month
Bio 6	Min Temperature of Coldest Month
Bio 7	Temperature Annual Range (BIO5-BIO6)
Bio 8	Mean Temperature of Wettest Quarter
Bio 9	Mean Temperature of Driest Quarter
Bio 10	Mean Temperature of Warmest Quarter
Bio 11	Mean Temperature of Coldest Quarter
Bio 12	Annual Precipitation
Bio 13	Precipitation of Wettest Month
Bio 14	Precipitation of Driest Month
Bio 15	Precipitation Seasonality (Coefficient of Variation)
Bio 16	Precipitation of Wettest Quarter
Bio 17	Precipitation of Driest Quarter
Bio 18	Precipitation of Warmest Quarter
Bio 19	Precipitation of Coldest Quarter

For species distribution, Maxent 3.3.3k software was used (Philips *et al.*, 2006). A total of 42 occurrence locality records were used to develop the model. 25% of the occurrence data were set aside as test points, and 10.000 background points were used to determine the distribution. Additionally, the regularization multiplier = 0.5, maximum iterations = 500 and convergence threshold = 10-5 were chosen in Maxent. In order to test the variable importance, the jackknife test of variable importance was chosen in Maxent, and the model was run as ten replicates. The result of the receiver operating characteristic (ROC) curve is important for model sensitization and the value of the area under the curve (AUC) closest to 1 indicated the best model

performance. A value near 0.5 suggests that the result is not better than random (Raes and ter Steege, 2007; Gallien *et al.*, 2012).

Results

We added new geopoint record from eastern black sea region that were not given in the literature. Bayburt is new provinces for distribution of *M. caspica* in Turkey. According to our constructed model, the distribution of species was generally observed to be between the northwestern Black sea region and the northern of the central Anatolia region, between eastern Anatolia and southern Anatolia regions. In addition, our model shows that the species may be distributed in low level in the other parts of Turkey indicated whitish or grayish color in the distributional map (Figure 1). Furthermore, the habitat preferences of the species were consistent with its distributional area (Figure 2). The environmental variables contributed to highest gain for distribution of *M. caspica* were elevation (21.4%), Bio 12 (Annual Precipitation, 20.5%), Bio 10 (Mean Temperature of Warmest Quarter, 14.8%) and Bio 17 (Precipitation of Driest Quarter, 11.6%). Other variables showed a lower contribution than percentage 35 (Table 3). Furthermore, elevation data presented the most useful information when used by itself in the Maxent analyze. According to the jackknife of regularized training gain of the species, the performance of the model was the lowest without elevation (Figure 3). All bioclimatic variables used in the present study are given in Table 3 with the contribution values. The final constructed models resulted in good and high AUC with a low standard deviation (0.912 ± 0.082) for *M. caspica*. The high AUC result and lower standard deviation exhibit that potential distribution area of the species was congruent with its habitat preferences inhabited localities.

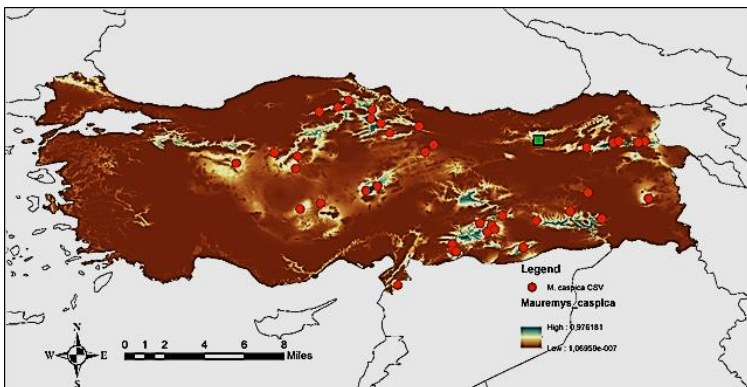


Fig 1: The expected potential distribution of *M. caspica* based on elevation and climatic conditions

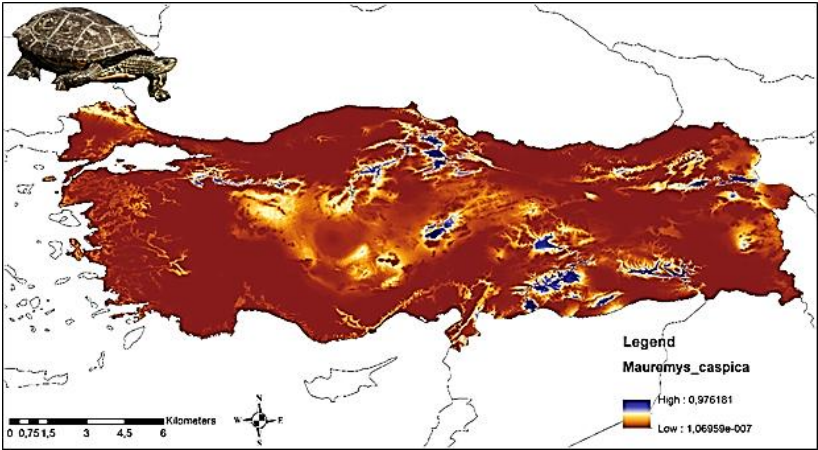


Fig 2: Habitat preferences of *M. caspica* in Turkey. Blue indicating suitable habitat and claret red representing unsuitable habitat

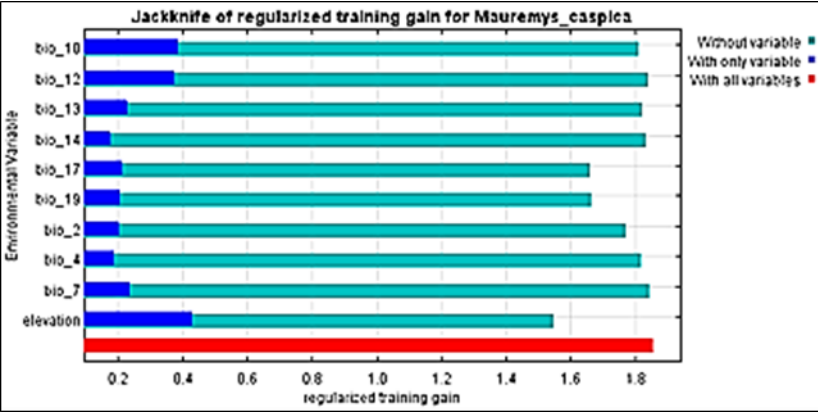


Fig 3: The results of the jackknife test of variable importance used in the model.

Table 3: Contributions of the climatic conditions to distribution modeling of *M. caspica*

S. No.	Variables	Contributions (%)
1	elevation	21.4
2	Bio 12	20.5
3	Bio 10	14.8
4	Bio 17	11.6
5	Bio 19	8.0
6	Bio 7	7.5
7	Bio 13	5.8

8	Bio 4	4.3
9	Bio 2	3.5
10	Bio 14	2.7

Discussion

M. caspica had a disconnected distribution in Turkey, according to all locality records (in the present study and literature), because the localities from central Anatolia and Northern Black sea region were separated the localities from eastern and southern Anatolia. Consistent with this data, the distributional model in the present study showed that northern Black sea region and southern Anatolia were highly suitable habitats for *M. caspica*. The species was also recorded in some localities from central Anatolia (Teynie, 1991; Fritz and Freytag, 1993; Schweiger 1994; Taşkavak *et al.* 1998; Ayaz *et al.* 2006; Fritz *et al.* 2008). However, the suitability of these localities in the distribution model was found lower in the present study.

M. caspica had very scattered populations in its distribution area in the Anatolia both literature and our constricted model (Teynie, 1991; Fritz and Freytag, 1993; Schweiger 1994; Taşkavak *et al.* 1998; Ayaz *et al.* 2008; Fritz *et al.* 2008). This characteristic has presented that the species will be harmonized with different habitat structures and climatic variables. Consistent with this situation, ecological niche model results in the present study showed that different climatic variables (temperature, precipitation, drought, etc.) contribute to the distribution of the species in Turkey.

Geographical distribution has an importance to contribute the conservation, ecology and evolution of the species. It is an important tool to get an answer to some questions such as where are geographic limitations in the distribution of species. According to literature, distribution of the *M. caspica* is mainly limited with the drainage system of the Euphrates, Tigris rivers, and Lake Van (Fritz and Freytag, 1993). Also, the species was known from the other river system such as Red river from central Anatolia and northern Black sea region (Teynie, 1991; Fritz and Freytag, 1993; Schweiger 1994; Ayaz *et al.* 2008; Fritz *et al.* 2008; Taşkavak *et al.* 1998). Moreover, any data and locality record were not known from eastern Black sea region in the literature for the species. In the present study, we provide for the first time new distributional area and new river system (Çoruh River) from eastern Black sea region.

According to the phylogeographic data, two species (*M. caspica* and *M. rivulata*) diverged in western Asia and that Europe and Central Anatolia, regions with low nuclear DNA diversity, were colonized from a south-eastern

or eastern radiation center (Fritz *et al.*, 2008). Besides, both phylogenetic relationships and fossil records shows stripe-necked terrapins was existed before Holocene (Lattin, 1967; Chkhikvadze, 1983; Chkhikvadze, 1989; Tuniyev, 1990; Tuniyev 1995; Myers *et al.*, 2000; Fritz *et al.*, 2008). Furthermore, all Anatolian individual of the *M. caspica* were phylogenetically within a single group (Fritz *et al.*, 2008). This condition can show that there is geographic isolation region between Anatolia and the other distributional region for the *M. caspica*. To better explain this situation, the habitats used by the species and the ecological niche requirements in these habitats should be needed well known. Accordingly, this study presented the habitats that the species could use in Anatolia and all ecological requirements in these habitats revealed the potential spread of the species and the abiotic factors affecting this distribution.

Many patterns in ecology, evolution and conservation are related to the geographical distribution of species, and these distributions depend on several factors that set range limits of species. In fact, these factors show how, in the course of the evolutionary process, geographic events in the areas where species live have changed the ecological processes. In the present study, variables affecting the distribution of the *M. caspica* emerged as a result of these processes. Elevation, rainfall and temperature are compatible with the current shape of the habitats of this species.

As a conclusion, species distribution modeling represents a scientifically important area. This study is very important because it is the first study that demonstrates the potential spread of the *M. caspica* using environmental variables. Moreover, this study is very important in terms of giving the first record of the species from the eastern Black Sea region except for the known distribution in Turkey.

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